

PROFESSIONAL SUMMARY

Policy & Computational Social Science researcher with expertise in program evaluation, causal inference, and applied machine learning, specializing in rigorous quasi-experimental and experimental methods including propensity score matching, difference-in-differences, regression discontinuity, and instrumental variables. Experienced in designing and executing end-to-end research and evaluation projects using large-scale administrative, survey, and behavioral datasets. Proven ability to translate complex quantitative findings into clear, actionable insights for policymakers & business leaders.

EDUCATION

COLUMBIA UNIVERSITY, School of International and Public Affairs (SIPA) New York, USA
Master of Public Policy & Administration, Data Science for Policy (*Program Evaluation & Social Policy*) May 2026
Relevant Coursework: Causal Inference & Program Evaluation, Applied Econometrics, Time Series Analysis, Applied Machine Learning, Micro- & Macro- Economics.

KING FAHD UNIVERSITY OF PETROLEUM & MINERALS, Engineering College Dhahran, Saudi Arabia
Bachelor of Science, Electrical Engineering (*Signal Processing & Artificial Intelligence Minors*) September 2021

RESEARCH EXPERIENCE

INTERNATIONAL LABOUR ORGANIZATION (ILO) New York, USA
Graduate Researcher November 2025 - Present

- Conducted institutional and legal analysis of national forced labour frameworks, mapping enforcement structures and governance mechanisms to ground empirical findings in real-world policy and regulatory context.
- Translated the ILO's legal definition of forced labour into a structured measurement framework, aligning international standards with nationally representative microdata to produce a scalable foundation for empirical policy analysis.
- Designed and implemented econometric strategies, including propensity score matching and logistic regression, using nearly 70,000 individual-level survey observations to estimate associations between forced labour and socioeconomic vulnerability factors.
- Designed a victim typology framework using PAM clustering with Gower distance similarity and MDS embedding to segment forced labour-affected individuals into distinct profiles defined by demographic, economic, and behavioral characteristics, producing classification outputs to support evidence-based policy targeting.
- Built a random forest model to predict the likelihood of individuals being in forced labour based on victim profiles and forced labour indicators.
- Contribute to research briefs and technical documentation, translating statistical findings into policy-relevant insights.

PROFESSIONAL EXPERIENCE

COMMUNICATION, SPACE & TECHNOLOGY COMMISSION (CST) Riyadh, Saudi Arabia
Cloud Computing Development Specialist April 2023 - August 2024

- Designed and implemented a comprehensive assessment framework for 32 cloud service providers, and developed a performance index that informed \$30M in national investment policies, enhanced market transparency, and fostered competition by incentivizing providers to improve quality and efficiency.
- Built interactive dashboards for senior leadership to track cloud provider performance in real time, enabling faster, data-driven strategic decisions and reducing average decision-making time by 60%.
- Produced analytical reports analyzing 1,200 organizations across 9 industries, identifying performance gaps and market demand trends to guide strategic growth and optimize investment decisions.
- Collaborated with policy, engineering, and investment teams to translate complex analyses into actionable strategies, driving 5 regulatory reforms in the national cloud regulatory framework.

DELOITTE Riyadh, Saudi Arabia
Business Analyst April 2022 - April 2023

- Assessed a financial-sector client's digital infrastructure, mapping services across 230 systems through surveys and interviews to identify gaps, mitigate risks, and address regulatory violations.
- Developed predictive models to estimate workload and resource demand, improving infrastructure planning accuracy and reducing potential system bottlenecks.
- Designed a hosting and data management strategy ensuring regulatory compliance and customer data protection while optimizing system load, improving overall system reliability by 40% and critical system reliability by 100%.

- Conducted a current-state assessment of a government agency, mapping services and digital infrastructure across 11 sub-agencies, identifying inefficiencies and driving the integration of 66 systems, reducing operational costs by 20% and improving infrastructure investment efficiency.
- Presented analytical findings and risk evaluations to senior leadership, shaping digital transformation decisions and informing long-term implementation roadmaps.

SAUDI TELECOM COMPANY (STC)

Access & Transport Strategy Developer

Riyadh, Saudi Arabia
December 2021 - April 2022

- Applied predictive and geospatial analysis to inform infrastructure investment decisions aligned with national connectivity and inclusion goals.
- Assessed regional disparities in service access to support equitable expansion planning.
- Developed a data-driven insights report to support strategic planning and align infrastructure decisions with national objectives

SAUDI ARAMCO, Ras Tanura, Saudi Arabia

Control and Network Engineer

September 2021 - December 2021

- Supported the migration and modernization of mission-critical industrial control systems underpinning large-scale energy export operations, including SCADA, PLC, and DCS architectures.
- Coordinated with cross-functional engineering and safety teams to align control systems with reliability, cybersecurity, and operational governance standards.

LEADERSHIP & ACTIVITIES

Program Assistant – Data Science For Policy Concentration at SIPA

Sep 2025 - Present

- Coordinate academic programming, workshops, and events bridging data science and public policy.
- Collaborate with faculty, administrators, and students to improve program operations and enhance student engagement.
- Support curriculum development and resources to strengthen applied data science training for policy students.
- Led hands-on workshops mentoring students in Python and R programming, data manipulation, and applied analytics.

SKILLS

- **Statistical & Econometric Methods:** OLS and generalized linear regression, logistic regression, fixed effects, difference-in-differences, instrumental variables (2SLS), regression discontinuity, propensity score matching, time series modeling and ARIMA-based forecasting, GARCH and VAR models.
- **Machine Learning:** Decision Trees, Random Forest, LASSO/Ridge regularization, k-Nearest Neighbor, PAM, and K-Modes clustering
- **Programming & Tools:** R (tidyverse, ggplot2, MatchIt, plm, ivreg, knitr, survey), Python (pandas, sklearn, statsmodels), Stata, SQL, GitHub, MS Office Suite (advanced).
- **Research & Communication:** client-facing report preparation, codebook documentation, and data visualization.